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**Achieving superior high-temperature strength–ductility synergy in a Cu–Ag–Zr alloy
via trace Ce doping induced atomic substitution and precipitate refinement**

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Abstract: High-conductivity copper alloys often suffer from severe degradation in strength and ductility at elevated temperatures, limiting their applications under extreme thermal conditions. Here, we demonstrate that the addition of a trace amount of Ce (0.2 wt.%) to a Cu–Ag–Zr alloy leads to exceptional high-temperature mechanical properties without significantly compromising thermal conductivity. Ce efficiently scavenges free oxygen and metallic impurities, leading to grain boundary purification and dispersion strengthening through nano-sized rare earth oxide particles. Consequently, this significantly improves the alloy's high-temperature strength and ductility while preserving its high thermal conductivity. Concurrently, the addition of Ce refines submicron secondary phase particles and increases their number density, resulting in a substantial decrease in the final grain size due to the recrystallization of PSN nuclei. Furthermore, Ce suppresses both discontinuous precipitation and coarsening of nanoscale Ag precipitates, thereby reinforcing the Zener pinning effect of the secondary phase on grain boundaries and inhibiting grain coarsening during recrystallization. These mechanisms work together to refine the grains, prevent grain coarsening during heating, and enhance the high-temperature mechanical properties of the alloy. This work establishes a viable strategy for optimizing the high-temperature performance of Cu–Ag–Zr alloys without compromising thermal conductivity, offering direct guidance for their application in high temperature service environments.

Key words: copper alloy; rare earth; second phase particle; thermal conductivity; high-temperature strength and plasticity; thermal stability; in-situ high temperature SEM/EBSD

1. Introduction

Advanced thermal management systems, such as the combustion chambers in high-thrust liquid rocket engines, present a formidable materials challenge: they demand structural materials capable of withstanding extreme temperatures while efficiently dissipating immense heat fluxes[1-3]. Key components like the inner wall of high-thrust liquid rocket engine combustion chambers require copper alloys that not only possess outstanding thermal conductivity to manage extreme heat fluxes from 3000–4000 °C gases but also retain excellent mechanical strength and ductility at elevated temperatures[4]. This combination of properties is critical for ensuring structural integrity and operational reliability under extreme thermal–mechanical cycling. Alloys, such as NARloy-Z (Cu–3Ag–0.5Zr), have been established as benchmark materials for their reliable performance [5-8]. However, the emergence of next-generation alloys such as GRCop-84 highlights the continuous quest for enhanced high-temperature capabilities [9-13]. A central challenge in this endeavor lies in overcoming the inherent trade-off between high thermal conductivity, which necessitates a pure copper matrix, and high-temperature strength, which typically requires the introduction of thermally stable second phase particles [14-18].

To strengthen copper alloys effectively, a widely adopted approach involves incorporating hard and insoluble second phase particles [14, 16, 18-23]. Although the addition of a second phase (e.g., intermetallic compounds, ceramic particles) to reinforce the copper matrix also results in a reduced thermal conductivity, this reduction is significantly less pronounced compared to when alloying elements are dissolved in the solid solution state. The stability of the microstructure and mechanical properties in Cu–Ag–Zr alloys at high temperatures primarily relies on the morphology, size and volume fraction of the Cu₄AgZr particle phases at micron/sub-micron levels and nanoscale Ag

precipitates [2, 7, 8]. Micron/submicron Cu₄AgZr particles primarily serve to pin dislocations, restrict grain boundary movement under high-temperature loading, and impede grain growth [12]. The spherical shape, more dispersed and denser distribution, and smaller size of Cu₄AgZr particles generally enhance the high-temperature strength and ductility of the alloy [24, 25]. Ag precipitation occurs in two ways: one involves spherical particles dispersed within the grains at around 10 nanometers, known as continuous/coherent precipitation; the other consists of non-spherical particles ranging from tens to hundreds of nanometers at grain boundaries, termed as discontinuous/incoherent precipitation [19, 26-29]. The discontinuous/incoherent precipitation of Ag helps in raising the recrystallization temperature and improving high-temperature mechanical properties, while also preserving the excellent electrical and thermal conductivity of the Cu–Ag–Zr alloys due to the coherent phase interface [5, 6, 26]. However, upon prolonged exposure to temperatures exceeding 500 °C, these continuous/coherent Ag precipitates become unstable [8, 30]. Accelerated elemental diffusion, particularly of Ag along grain boundaries, triggers coarsening of precipitates and a transition to discontinuous precipitation [12]. This microstructural degradation leads to a pronounced decline in high-temperature strength, ductility, and thermal conductivity. Therefore, suppressing elemental diffusion and the concomitant coarsening of Ag is paramount for enhancing high-temperature performance of the Cu–Ag–Zr alloys.

Metals primarily rely on electrons for thermal conduction. Introducing impurity atoms (e.g., Ca, S, O) into a copper matrix induces lattice distortion, which enhances electron scattering, reduces the electron mean free path, and consequently decreases the alloy's thermal conductivity [9, 10, 31]. Impurity elements in copper alloys are primarily categorized into non-metallic elements (such as oxygen) and metallic elements (such as calcium), originating from raw materials, processing

procedures, and environmental contamination [32-34]. These impurities can significantly impair high-temperature mechanical properties and thermal conductivity by disrupting the integrity of the crystal lattice and degrading the characteristics of grain boundaries [35]. For example, oxygen (O) readily forms Cu_2O inclusions with copper, which act as strong electron scattering centers, thereby reducing thermal conductivity and weakening the cohesion of grain boundaries, thus promoting intergranular fracture at elevated temperatures [4, 5]. Calcium (Ca) has a large atomic radius, which induces significant lattice distortion upon solid solution and results in a severe decrease in thermal conductivity. Additionally, the segregation of CaO at grain boundaries results in stress concentration at high-temperature interfaces, ultimately compromising high-temperature strength [13].

Rare earth (RE) element microalloying presents a promising strategy to address this multifaceted challenge. In copper alloys, the effect of RE elements has been attributed to both purification and microalloying [21, 22]. Their high chemical reactivity enables them to scavenge detrimental impurities such as oxygen, sulfur, and calcium from the copper matrix and grain boundaries [36, 37]. This purification effect enhances both thermal conductivity and high-temperature ductility. Simultaneously, RE elements can interact with alloying components to form fine, dispersed second phase particles with high melting points [38]. These particles effectively pin grain boundaries and dislocations, thereby improving resistance to recrystallization and grain growth, which stabilizes strength and ductility at high temperatures.

Cerium (Ce) is the most abundant light rare-earth element (comprising ~60% of total rare-earth reserves), featuring low cost and well-established industrial refining processes—thereby circumventing the high costs and supply-chain vulnerabilities associated with scarce heavy rare-earth elements (e.g., Y, Gd) and enabling scalable production of high-temperature Cu alloys. The

grain-refining effect of Ce is negligible at concentrations ≤ 0.05 wt.%, whereas additions of 0.05–0.2 wt.% Ce significantly refine the grain structure of Cu–Ag alloys, with tensile strength attaining a maximum at 0.2 wt.% Ce [34, 35]. Besides, the addition of small amounts of Ce has been shown to enhance the room-temperature tensile strength of Cu alloys without significantly compromising their electrical conductivity [34, 39]. However, the effects of rare earth element incorporation, particularly Ce, on the high-temperature physical and mechanical properties of copper-based alloys remain poorly understood. Given the demanding thermo-mechanical environment experienced by combustion chamber walls, it is essential to systematically investigate how Ce doping influences both the thermal conductivity and high-temperature mechanical behavior of the Cu–Ag–Zr alloy. By utilizing powder metallurgy to introduce trace amounts of Ce, the high-temperature thermal stability of the Cu–3Ag–0.5Zr alloy is successfully improved while maintaining its exceptional thermal conductivity in this study. The addition of trace Ce efficiently eliminates the residual impurities from the matrix, increases the population of submicron Cu₄AgZr particles and refines them at the same time. Additionally, it changes the precipitation pattern of the Ag phase during aging, causing the development of fine, dispersed, and cohesive precipitates throughout the material. Through a combination of atomic-resolution microscopy and in-situ high-temperature SEM/EBSD analysis, we provide direct evidence of how trace Ce doping tailors the precipitation behavior and grain stability, providing insights for designing high-performance copper alloys for extreme environments.

2. Experiments

2.1 Sample preparation

In this study, 0.2 wt.% of the rare earth element Ce was added to the NARloy-Z (Cu–3Ag–

0.5Zr) alloy. The raw materials used were high-purity electrolytic copper (purity greater than 99.9%), high-purity silver (purity greater than 99.9%), a Cu-50% Zr (mass fraction, the same below) master alloy, and a Cu-10% Ce master alloy. Because of the extremely low solid solubility of Zr and Ce in Cu, macrosegregation in the alloy cannot be avoided with conventional casting methods [2, 24]. To produce alloys with a fine grain size and dispersed second-phase particles, the alloy is fabricated using gas atomization and powder metallurgy techniques. Pre-alloyed powders of two compositions were prepared using vacuum melting and argon atomization. The raw materials were formulated according to the compositions of the Cu-3Ag-0.5Zr alloy and the Cu-3Ag-0.5Zr-0.2Ce alloy, respectively. The melting and atomization chambers of the vacuum atomization apparatus were evacuated to 1.0×10^{-2} Pa and maintained for 5 min. After ensuring that the vacuum level did not drop, argon was then filled to standard atmospheric pressure as a shielding gas. A high-frequency induction device was then activated for heating. Once the alloy in the crucible was completely melted, the melt temperature was raised to 1350 °C and held for 30 min in preparation for atomization and powder production. Atomization and pulverization were initiated by adjusting the argon atomization pressure to 3.5 MPa. The collected powder was sieved through a 200-mesh sieve, immediately vacuum-packaged, and stored in a vacuum drying oven. The actual composition of the two sets of pre-alloyed powders was determined using a SPECTRO BLUE SOP inductively coupled plasma (ICP) spectrometer. The results are shown in Table 1. The resulting powders were densified using hot isostatic pressing. The powders were first loaded into a steel cylindrical can with internal dimensions of 120 mm $\times$ 150 mm. The tap density of the pre-alloyed powder for each component ranged from 5.3 to 5.6 g $\cdot$ cm $^{-3}$. The sealed can was then placed in a hot isostatic pressing furnace for pressurization and heating. The hot isostatic pressing process was held at a holding pressure of 150

MPa and a holding temperature of 830 °C. The pressing and heating processes were synchronized, and the holding time was 2 h. The hot isostatically pressed alloys were then cold rolled. The room temperature cold-rolled samples measured 120 mm × 40 mm × 12 mm, with a total deformation of 80% and a deformation of 10% per pass. The rolled samples were then aged at 400 °C for 30 min. Aging at 400 °C for 30 min for the rolled Cu-Ag-Zr alloy had been demonstrated in previous studies to notably enhance ductility, recover electrical conductivity, and preserve strength [2].

Table 1 Composition of the Ce-free and Ce-containing alloys (wt.%).

Alloy	Ag	Zr	Ce	Al	Bi	Fe	S	Mn	Sn	Si	Cu
1#	3.07	0.48	--	0.009	0.002	0.007	0.001	0.001	0.002	0.001	Balance
2#	2.94	0.53	0.22	0.008	0.001	0.003	0.002	0.001	0.001	0.005	Balance

2.2 Materials characterization

The morphology of the precipitated phase was analyzed using a field-emission scanning electron microscope (SEM, CIQTEK SEM5000). Mechanically ground samples were electropolished at -25 °C using a 20% volume fraction nitric acid + 80% volume fraction methanol solution at a voltage of 10 V. Fine microstructural characterization was performed using a transmission electron microscope (TEM, Thermo Scientific Spectra Ultra S/TEM) at an accelerating voltage of 300 kV. Thin foils for TEM observation were mechanically thinned to 30-40 μm and then finally thinned using a Gatan PIPS 695 ion miller. Crystal structures of the aged alloy were characterized using X-ray diffraction (XRD, Rigaku NANOPIX-WE) with Mo K_α radiation ($\lambda = 0.071$ nm). Thin foil samples were mechanically polished to a thickness of approximately 100 μm prior to XRD measurements. High-temperature tensile testing was performed on a microcomputer-controlled electronic universal testing machine (KAIPLE-P5) equipped with a high-temperature furnace and a video extensometer measurement system (LVE-5M), enabling high-temperature

tensile testing and obtaining stress-strain curves. For both room-temperature and high-temperature tensile samples, dog-bone specimens with a gauge length of $3\text{ mm} \times 1.5\text{ mm} \times 1\text{ mm}$ were cut along the rolling direction. Tensile testing was performed at a displacement rate of 0.2 mm/min , with the tensile direction parallel to the rolling direction. High-temperature tensile testing temperatures were $350\text{ }^{\circ}\text{C}$, $400\text{ }^{\circ}\text{C}$, and $500\text{ }^{\circ}\text{C}$, respectively. The heating rate for the high-temperature tensile tests was approximately $15\text{ }^{\circ}\text{C/min}$, and the tensile test was initiated after a 10 min hold at the test temperature. Conventional large-sized samples were avoided for high-temperature tensile tests for two key reasons. Small samples enabled rapid, uniform heating of the gauge length with minimal temperature variation, matching the rapid heating of combustion chamber inner wall materials in service. Additionally, large samples brought high costs for high-temperature mechanical extensometer, and their large gauge lengths also caused video extensometer markers to move out of the camera view (due to high plasticity), leading to test failure. While small samples allowed precise stress-strain measurement via video extensometer.

2.3 *In-situ* SEM and EBSD tests

In-situ SEM heating tests were conducted under scanning electron microscopy to investigate the impact of trace Ce on the precipitation and growth behavior of the second phase at elevated temperatures ($600\text{ }^{\circ}\text{C}$) using a ZepTools PicoFemto-SEP SEM5000N stage. Fig. S1a displays the in-situ heating device and the sample. The dimensions of the sample are $2.5\text{ mm} \times 8\text{ mm}$, with a thickness of 0.8 mm . After mechanical polishing, the sample was then treated with electrochemical polishing. The heating curve and the acquisition time window of the SEM data are displayed in Fig. S1b. To avoid any major alterations in the microstructure of the sample, the data collection time is restricted to a maximum of 5 min. During the insulation process, the temperature fluctuation is kept

within ± 1 °C. Both samples were heated and kept at a constant temperature following the heating curve, and the time windows for data collection remained consistent. Data were first collected at room temperature, as shown in Fig. S1b. Subsequently, the sample was incubated at 600 °C for 30 min before the second data collection. For the third data collection, the sample was incubated continuously for 60 min. Finally, before the fourth data collection, the sample was incubated continuously for 120 min. The SEM data collected at each stage were labeled according to temperature and continuous incubation time.

In order to analyze the effect of rare earth Ce addition on the thermal stability of alloys more intuitively, the recrystallization and growth behavior of grains in two alloys during heating at 500-600 °C were observed through in-situ heating EBSD experiments using a ZepTools MHE stage. Figure S2a shows the in-situ chip and its heating device. Figure S2b shows the SEM image of the test sample welded in the heating zone of the chip. The sample is prepared using FIB, with a test surface size of $30\text{ }\mu\text{m} \times 25\text{ }\mu\text{m}$ and a thickness of $3\text{ }\mu\text{m}$. Figure S2c shows the testing process of in-situ heating of the sample in a scanning electron microscope. The heating curve of the sample and the EBSD data acquisition time window are shown in Fig. S2d. To avoid any major alterations in the microstructure of the sample, EBSD data collection time is restricted to a maximum of 8 min. During the insulation process, the temperature fluctuation is kept within ± 1 °C. Both samples were heated and kept at a constant temperature following the heating curve, and the time windows for data collection remained consistent. The data were initially collected at room temperature, as shown in Fig. S2d. After that, the sample was incubated at 500 °C for 50 min before the second data collection. Subsequently, the sample was continuously incubated for 30 min for the third data collection, 60 min for the fourth data collection, 240 min for the fifth data collection, and 270 min

(including 13 min at 600 °C) for the sixth data collection. Finally, before the seventh data collection, the sample was continuously incubated for 330 min (including 73 min at 600 °C). The EBSD data collected at each stage are named based on temperature and continuous holding time.

3. Results

3.1 Mechanical properties at room temperature

To investigate the impact of Ce on the thermal stability of the alloy, the tensile properties and hardness of the Ce-free alloy and Ce-containing alloy after annealing at high temperatures are compared, as presented in Fig. 1 and Table 2. As shown in Fig. 1a-c, there is a 3.7% decrease in tensile strength of the rolled Ce-free alloy after annealing at 350 °C for 1 h. The drop in tensile strength of the Ce-containing alloy is 3.4%, showing a minor difference compared to the alloy without Ce. However, as the annealing temperature increases, the difference in the strength decrease of the two alloys after annealing will become more apparent. The tensile strength of the Ce-free alloy decreases by 15.7% after annealing at 400 °C for 1 h, while the Ce-containing alloy shows an 11.4% reduction at the same temperature. At 450 °C, the tensile strength of Ce-free alloy decreases by 24.5%, compared to a 17.8% decrease in the Ce-containing alloy. Additionally, the hardness values of the Ce-containing alloy decrease at a slower rate compared to the Ce-free alloy with increasing annealing temperature. It is also worth mentioning that the presence of Ce will result in a slight decrease (4% - 6%) in the elongation of the annealed alloy at various temperatures.

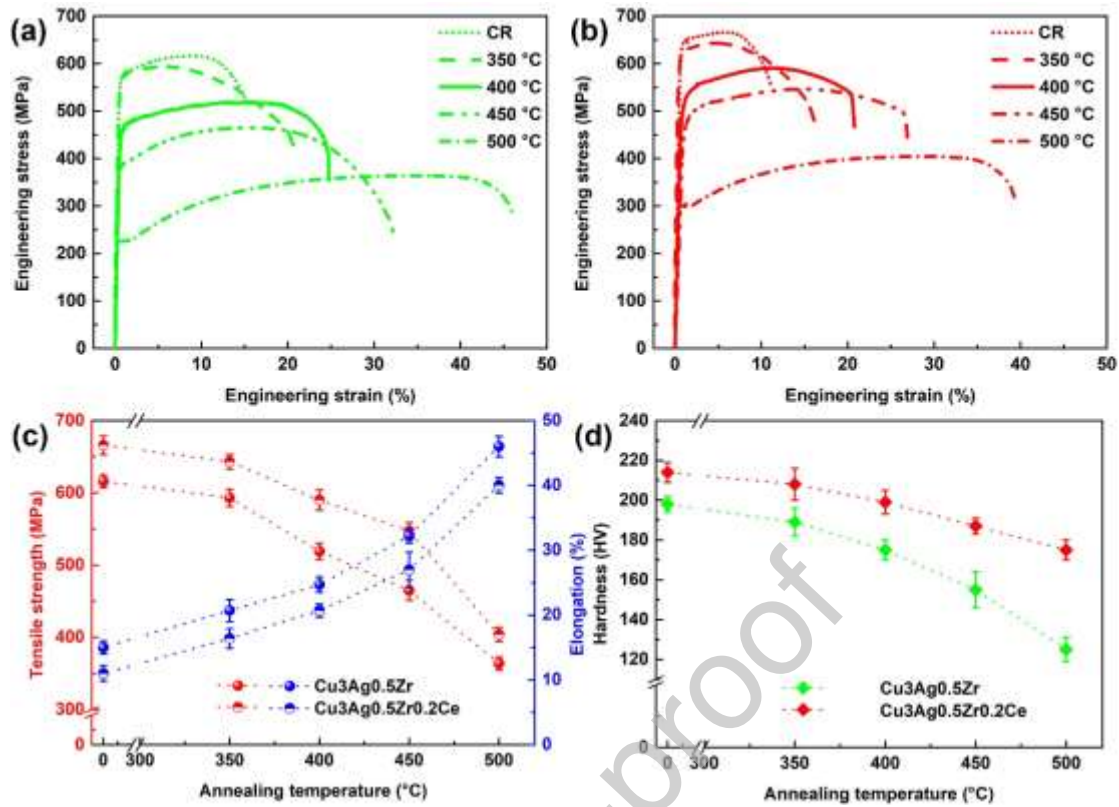


Fig. 1 Mechanical properties at room temperature of the alloys after annealing for 1 h at different temperatures. Engineering stress-strain curves of the (a) Ce-free and (b) Ce-containing alloys. (c) Tensile properties and (d) hardness values with the increase of annealing temperature.

Table 2 Comparison in room-temperature mechanical properties of the Ce-free and Ce-containing alloys after annealing at different temperatures for 1 h, respectively.

Materials	Annealing temperature (°C)	Tensile strength (MPa)	Elongation (%)	Hardness (HV)
Ce-free alloy	CR	616 ± 9.1	15 ± 1.0	198 ± 4.2
	350	593 ± 12.3	21 ± 1.7	189 ± 7.3
	400	519 ± 11.4	25 ± 1.2	175 ± 5.4
	450	465 ± 13.6	32 ± 1.1	155 ± 9.2
	500	364 ± 9.4	46 ± 1.6	125 ± 5.8
Ce-containing alloy	CR	666 ± 13.4	11 ± 1.2	214 ± 4.7
	350	643 ± 10.1	16 ± 1.5	208 ± 7.9
	400	590 ± 14.0	21 ± 1.0	199 ± 6.2
	450	547 ± 11.6	27 ± 2.7	187 ± 4.3
	500	404 ± 9.8	40 ± 1.2	175 ± 5.4

3.2 Temperature dependence of mechanical and physical properties

The thermal conductivity and tensile properties of the alloys at high temperatures are shown in Fig. 2a, b and Table 3. As shown in Fig. 2a, it can be observed that the thermal conductivity of both alloys initially rises and then falls as the temperature increases, peaking at 300 °C. The incorporation of trace rare earth Ce leads to a slight decrease in thermal conductivity. At room temperature, the thermal conductivity of the Ce-free alloy is $11 \text{ W}\cdot\text{m}^{-1}\text{K}^{-1}$ higher than that of the Ce-containing alloy. As the test temperature increases, the difference in thermal conductivity between the two alloys gradually decreases. At 500 °C, the thermal conductivity of the Ce-free alloy is $5 \text{ W}\cdot\text{m}^{-1}\text{K}^{-1}$ (1.5%) higher than that of the Ce-containing alloy. The good heat transfer ability of the alloy is obviously preserved after adding a small amount of Ce.

At 350 °C, the Ce-free alloy has a high-temperature strength of 512 MPa, while the Ce-containing alloy shows an increase of 65 MPa in strength without a significant decrease in elongation. At 400 °C, the Ce-containing alloy still has an advantage of nearly 20 MPa in high-temperature strength compared to the Ce-free alloy. When the testing temperature reaches 500 °C, the strength of the Ce-free alloy decreases to 239 MPa with an increase in plasticity to 32%. On the other hand, the Ce-containing alloy maintains a high strength of 263 MPa with good high-temperature plasticity of about 28%. As can be seen, adding a small amount of Ce to the Cu–Ag–Zr alloy can maintain high-temperature plasticity and thermal conductivity while enhancing high-temperature strength. When directly comparing the high-temperature tensile strength and elongation of our samples with other high-performance copper alloys typically utilized in high-temperature environments, as displayed in Fig. 2c, it is evident that our samples exhibit exceptional combinations of high-temperature strength and plasticity.

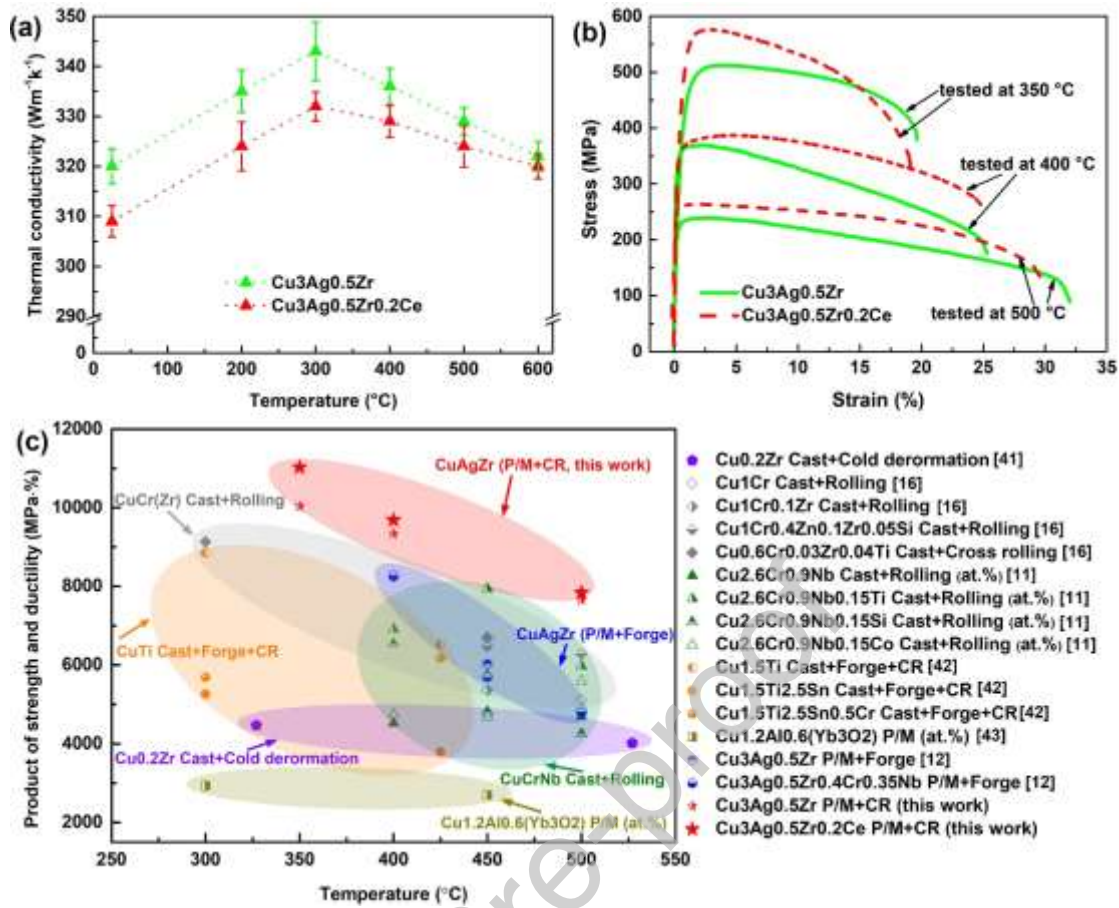


Fig. 2 (a) Variation of thermal conductivity as a function of temperature. (b) Stress-strain curves of the Ce-free and Ce-containing alloys tested at 350, 400, and 500 °C, respectively. (c) Comparison of product of strength and ductility at different temperatures with other copper alloys for high temperature applications[11, 12, 16, 40-43].

Table 3 Thermal conductivity and tensile properties of the alloys tested at high temperatures.

Materials	Test temperature (°C)	Tensile strength (MPa)	Elongation (%)	Thermal conductivity (W·m ⁻¹ ·K ⁻¹)
Ce-free alloy	25	--	--	320 ± 3.5
	200	--	--	335 ± 4.2
	300	--	--	343 ± 5.9
	350	512 ± 8.3	17 ± 3.1	--
	400	369 ± 11.8	25 ± 2.5	336 ± 3.6
	500	239 ± 9.7	32 ± 2.8	329 ± 2.7
	600	--	--	322 ± 3.0
Ce-containing alloy	25	--	--	309 ± 3.1
	200	--	--	325 ± 4.9

300	--	--	332 ± 2.9
350	577 ± 10.1	19 ± 2.7	--
400	387 ± 10.9	21 ± 2.2	330 ± 3.2
500	263 ± 13.2	28 ± 3.5	324 ± 4.1
600	--	--	320 ± 2.5

3.3 Microstructural features of precipitates

Figure 3 illustrates the microstructure of submicron second phase particles in the rolled Ce-free and Ce-containing alloys. SEM images reveal that the particle sizes are well-controlled in both alloys. However, adding trace amounts of the rare earth element Ce to the alloy results in a significantly higher concentration and more uniform distribution of submicron spherical second phase particles. EDS mappings show that the submicron second phase particles in the Ce-free alloy primarily consist of Cu, Ag, and Zr elements, while the those in the Ce-containing alloy also include Ce. The selected area electron diffraction (SAED) patterns of the second phase particles, acquired along three different low-index zone axes, indicate that this phase possesses a face-centered cubic (fcc) crystal structure with a lattice constant of $7.07 \text{ \AA}$, as shown in Fig. 3d₁, d₂, and d₃. This measured value is very close to the theoretical value of the lattice constant of the Cu₄AgZr phase (fcc structure, $a = b = c = 7.01 \text{ \AA}$)[44]. Notably, the Ce-rich second phase particle observed in Fig. 3f also exhibits an fcc crystal structure based on SAED patterns from the same zone axes, as depicted in Fig. 3h₁, h₂, and h₃. However, the lattice constant calculated for the Ce-rich second phase particle is $6.69 \text{ \AA}$, indicating a smaller unit cell compared to the Cu₄AgZr phase in the Ce-free alloy.

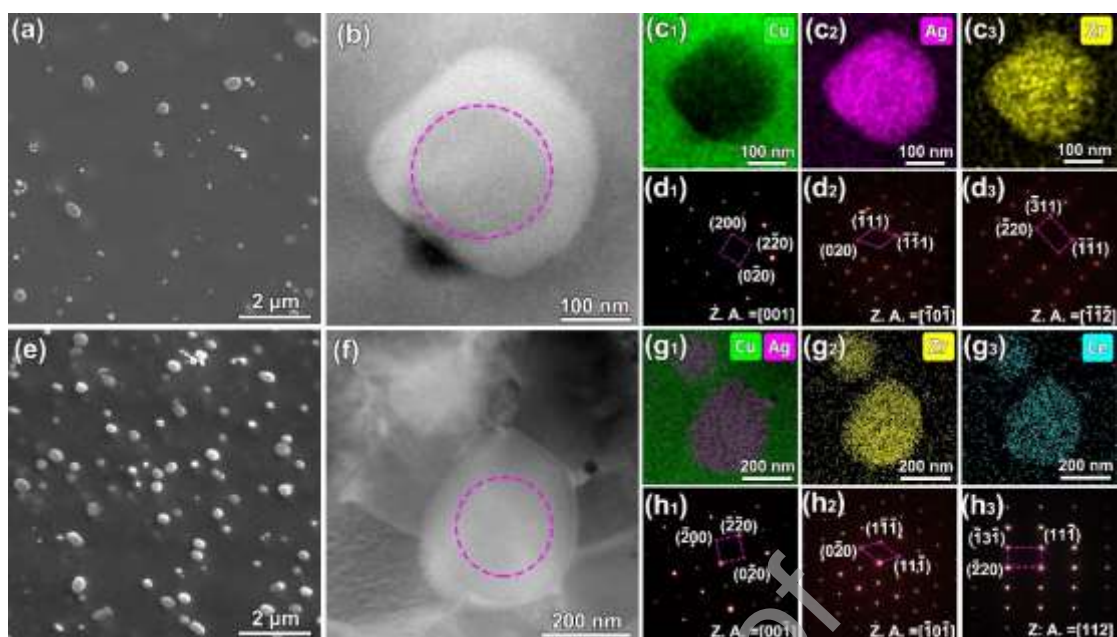


Fig. 3 Microstructures of the submicron second phase particles in the rolled Ce-free alloy and Ce-containing alloy. (a) SEM image, (b) STEM image, (c₁-c₃) EDS mappings, and (d₁-d₃) selected area electron diffraction (SAED) patterns along three different low-index zone axes taken from the dotted circles in figure 3(b). (e) SEM image, (f) STEM image, (g₁-g₃) EDS mappings, and (h₁-h₃) SAED patterns along three different low-index zone axes taken from the dotted circles in figure 3(f).

The atomic-scale spatial configuration of the Cu₄AgZr phase in the Ce-free alloy and the Ce-rich phase in the Ce-containing alloy is further examined, as illustrated in Fig. 4. The high-resolution STEM-HAADF images reveal that the atomic structures of the Cu₄AgZr phase and the Ce-rich phase remain unchanged when viewed along the [101] axis. The atomic positions of various elements follow a regular pattern in the Cu₄AgZr phase. Analysis using atomic scale EDS-mapping reveals that the positions of Cu and Ag atoms in the Ce-rich phase remain unchanged, while there is an overlap between the positions of Ce and Zr atoms. This implies that Ce atoms replace certain Zr atoms within the Cu₄AgZr phase, creating a competitive dynamic between the two types of atoms in the Ce-rich phase. Hence, in this research, we shall refer to this submicron second phase in the

Ce-containing alloy as $\text{Cu}_4\text{Ag}(\text{ZrCe})$.

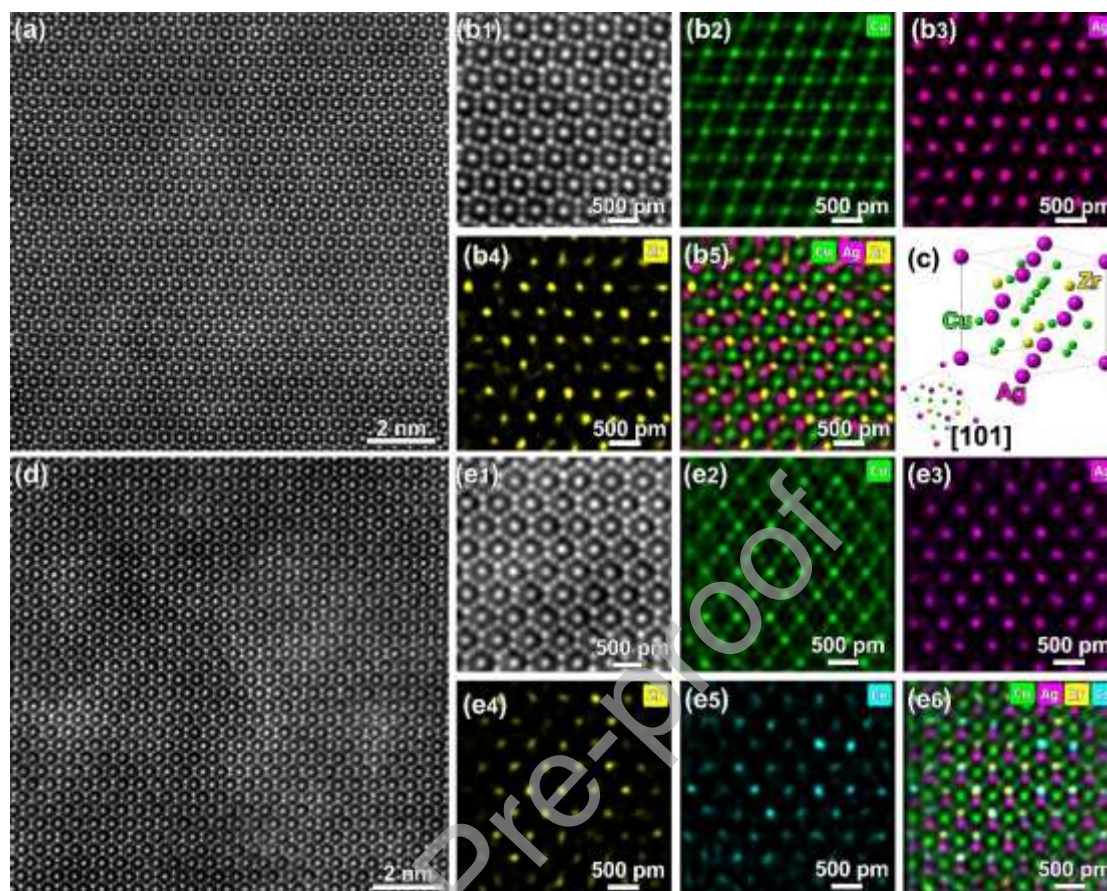


Fig. 4 (a-b₅) High-resolution HAADF-STEM image taken at [101] zone axis and corresponding atomic scale EDS mappings of Cu_4AgZr phase in the rolled Ce-free alloy. (c) The crystal structure model of the Cu_4AgZr phase. (d-e₆) High-resolution HAADF-STEM image and corresponding atomic scale EDS mappings of $\text{Cu}_4\text{Ag}(\text{ZrCe})$ phase in the rolled Ce-containing alloy. The comparative analysis of atomic scale EDS mappings of Cu_4AgZr and $\text{Cu}_4\text{Ag}(\text{ZrCe})$ phases revealed that Ce atoms occupy the positions of some Zr atoms.

Figure 5a exhibits a high density of nano-sized precipitates measuring between 50 nm and 100 nm in the aged Ce-free alloy. In contrast, figure 5b shows a lower density of newly formed precipitated phases ranging from tens to hundreds of nanometers in the aged Ce-containing alloy. The newly formed precipitates during aging in both alloys are further examined by TEM, as depicted

in Fig. 5c-d₃. Only a few number of Cu₄AgZr particles precipitate during aging in the Ce-free alloy, with the majority of the precipitates being composed of Ag according to the EDS results (Fig. 5c-c₃). These nanosized Ag precipitates indicated by magenta arrows in Fig. 5c are in line with the particles ranging from 50 nm to 100 nm seen in SEM in Fig. 5a. Ag precipitates with relatively large sizes are discovered both at grain boundaries and within the grain interior, as illustrated in Fig. 6. It is often difficult to sustain a coherent relationship between the precipitated phases on grain boundaries and the matrix. A high-resolution HAADF-STEM image (Fig. 6h) revealed that the Ag precipitates about 60 nm within the grain interior show an incoherent relationship with the Cu matrix. This behavior is characteristic of discontinuous precipitation of Ag [26, 45]. Figure 5d-d₃ shows that there are no Ag precipitates between 50 nm and 100 nm in the aged Ce-containing alloy. Instead, most of the newly formed particles ranging from tens to hundreds of nanometers are Cu₄Ag(ZrCe) particles. In the aged Ce-containing alloy, a high density of Ag precipitates smaller than 10 nm is observed, as depicted in Fig. 7. These precipitates mostly have a short rod-like shape, with the shorter dimension measuring only a few atomic layers. Additionally, a smaller population of larger precipitates, around 10 nm in size, appears nearly spherical, with increasing sphericity as the particle size grows. These size-dependent Ag precipitates are fully coherent with the matrix and are not present in the Ce-free alloy. The size distribution of the Ag precipitates and Cu₄AgZr/Cu₄Ag (ZrCe) phases in the aged alloy is statistically analyzed in Fig. 8. The statistics of the submicron Cu₄AgZr/Cu₄Ag (ZrCe) phase are based on 10 SEM images with magnification consistent with Fig. 5a, b. The statistics of the 50-100 nm Ag precipitates in the Ce-free alloy are based on 10 STEM images with magnification consistent with Fig. 5c. The statistics of the Ag precipitates below 10 nm in the Ce-containing alloy are based on 10 TEM high-resolution images with magnification

consistent with Fig. 7a. Table 4 exhibits the volume fraction and average size of various precipitated phases in the aged alloys.

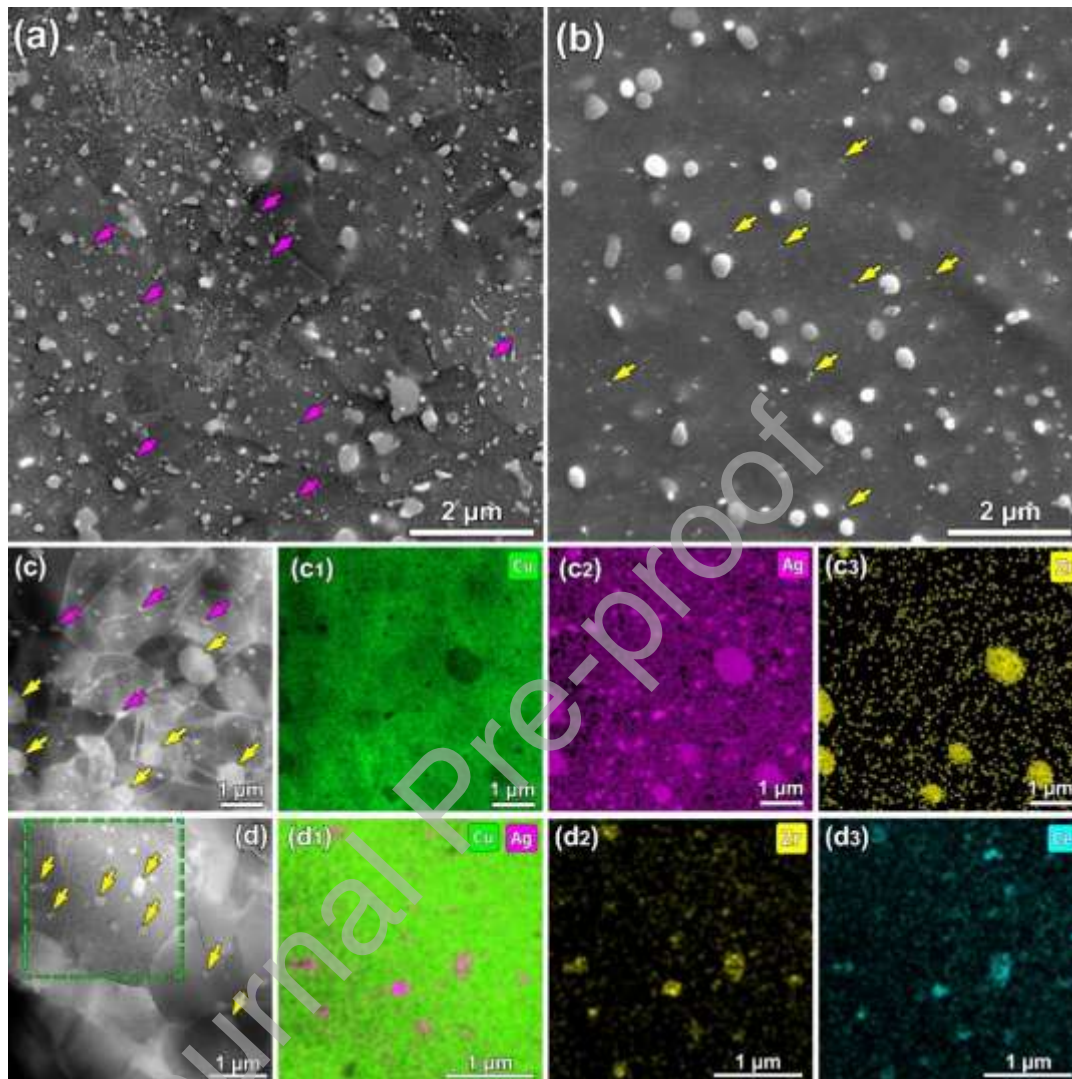


Fig. 5 SEM images of the nano-sized precipitates in the aged (a) Ce-free alloy and (b) Ce-containing alloy. STEM images and corresponding EDS mappings of the aged (c-c₃) Ce-free alloy and (d-d₃) Ce-containing alloy.

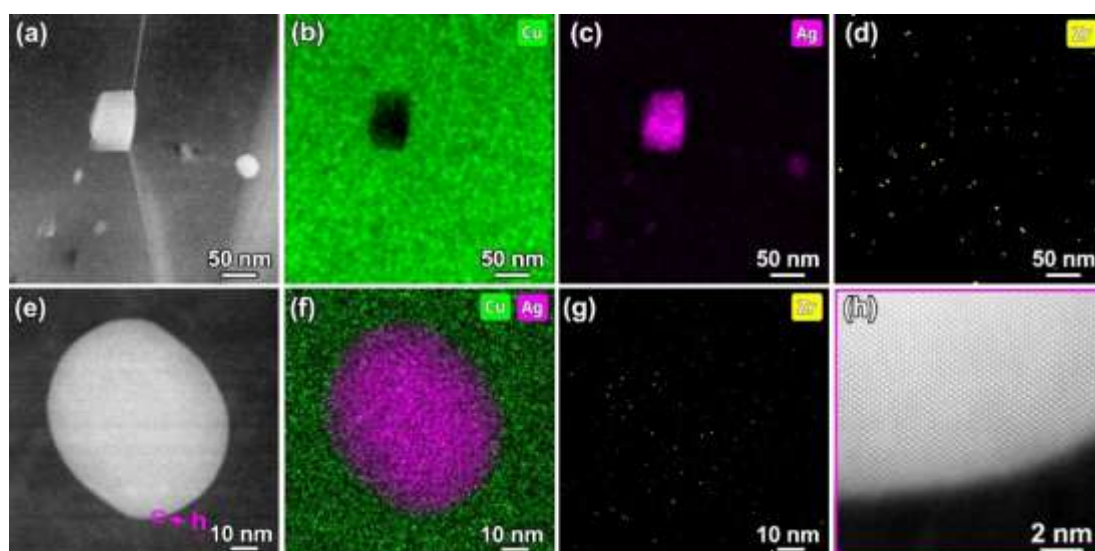


Fig. 6 (a, e) STEM images and (b-d, f, g) corresponding EDS maps of the incoherent Ag precipitates at grain boundaries and within grains in aged Ce-free alloy. (h) High-resolution HAADF-STEM image of the incoherent Ag phase taken from position h indicated in figure 6(e) viewed along the $[110]_{\text{Ag}}$.

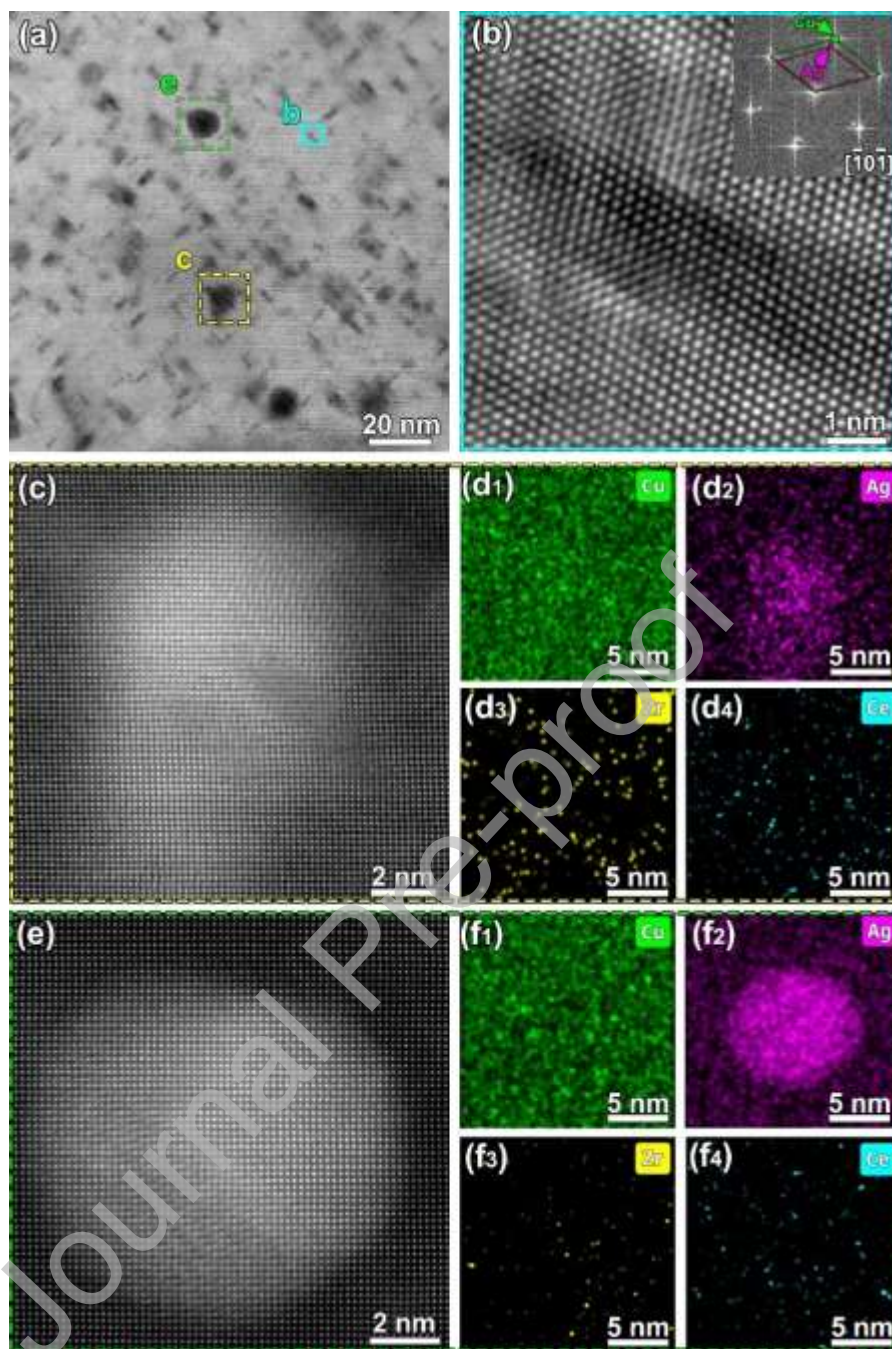


Fig. 7 (a) Coherent Ag precipitates with different size in aged Ce-containing alloy. (b) TEM bright field image of the short rod-like Ag precipitates with several atomic layers thick, the inserted image is the fast fourier transform (FFT) pattern of the short rod-like Ag precipitates. (c, e) HAADF-STEM images and (d₁-d₄, f₁-f₄) corresponding EDS maps of the spherical Ag precipitates with the size of about 10 nm.

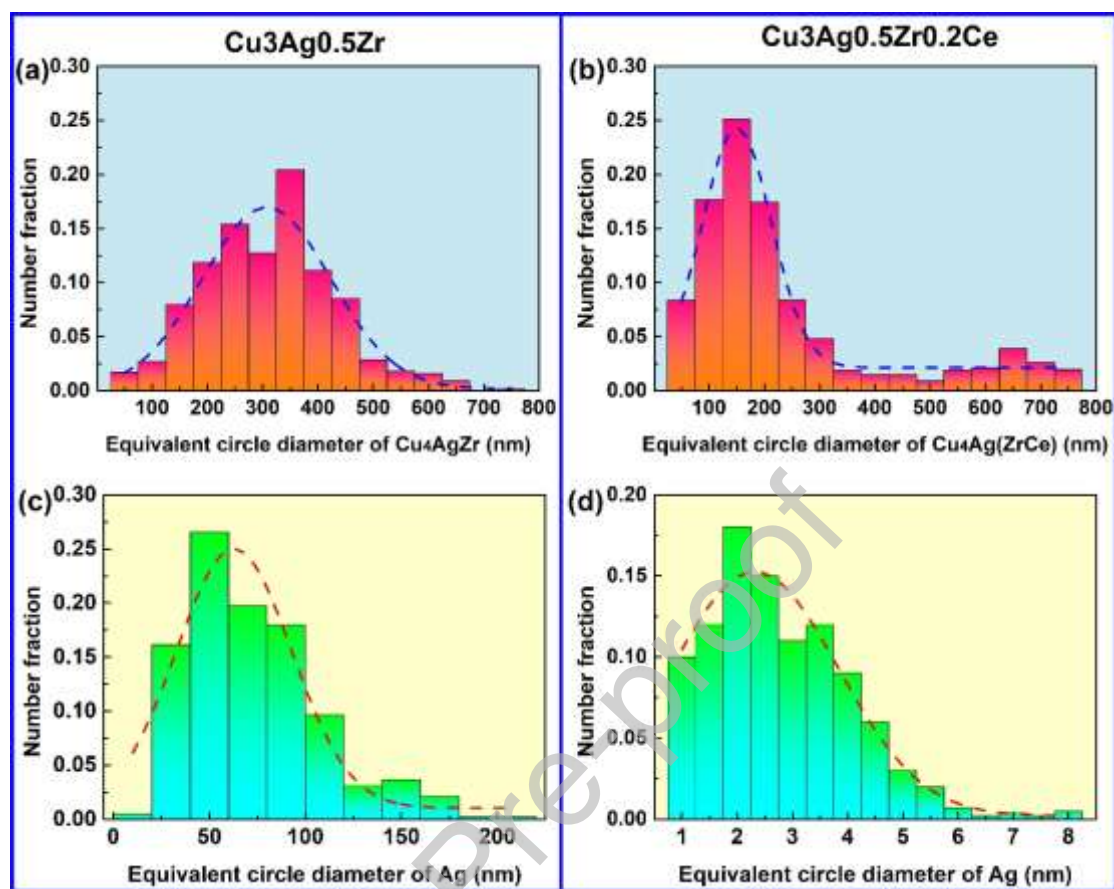


Fig. 8 Size distribution of second phase particles in the aged alloys

Table 4 Volume fraction and average size of the main second phase particles in the aged alloys.

Sample	Second phase particle	Volume fraction	average particle size
Cu-3Ag-0.5Zr	Ag	4.6%	73 ± 1.2 nm
	Cu ₄ AgZr	5.8%	312 ± 6.5 nm
Cu-3Ag-0.5Zr-0.2Ce	Ag	10.2%	3 ± 0.6 nm
	Cu ₄ Ag(ZrCe)	7.1%	235 ± 4.9 nm

Nano-sized rare earth oxides were observed at/near the grain boundaries in the aged Ce-containing alloy, as depicted in Fig. 9. Some slightly larger oxide particles were found to be rich in Zr and Ce, while some slightly smaller oxide particles also contained Ca in addition to Zr and Ce. In the HAADF-STEM mode, Ca-containing oxides appear as distinct core-shell structures, as shown in Fig. 10a-b₆. The core has a diameter of approximately 15 nm, with low calcium content but higher

oxygen and Ce content compared to the outer layer. The central location of cerium/zirconium oxide suggests that it precipitates first, and then absorbs Ca impurity elements in the Cu matrix. The standard Gibbs free energy of formation for zirconium oxide, cerium oxide, and calcium oxide was $-1000 \sim -1100$ kJ/mol, $-1026 \sim -1821$ kJ/mol, and $-550 \sim -640$ kJ/mol, respectively [46, 47]. At the same temperature, the formation of zirconium oxide and cerium oxide exhibits a lower Gibbs free energy than that of calcium oxide. When the local oxygen partial pressure is sufficient to initiate oxidation, Zr and Ce react with oxygen atoms to nucleate the nano-core preferentially, due to their high oxygen affinity, during alloy melting, atomization, sintering, and subsequent heat treatment. Upon reaching the surface of the Zr-Ce-O nano-core, CaO (lattice constant: $4.81 \text{ \AA}$) undergoes epitaxial growth on the Zr-Ce-O ($\text{Ce}_{0.5}\text{Zr}_{0.5}\text{O}_2$ lattice constant: $5.26 \text{ \AA}$), facilitated by a small lattice mismatch of approximately 8.5%, which is below the typical threshold for epitaxial growth ($<10\%$) [48, 49]. The key lies in the creation of calcium oxide on existing cerium zirconium oxide particles, acting as a substrate to lower the nucleation free energy barrier by reducing the interfacial mismatch, ultimately leading to the achievement of heterogeneous nucleation. The reduced interfacial energy promotes the layered, continuous deposition of a dense Ca-rich shell. Analogous mechanisms have been observed in Al-Ce/Zr-O and Y-Cr-O systems, suggesting broad applicability across multicomponent oxides, characterized by nucleation of high oxygen-affinity elements followed by encapsulation via low diffusion-barrier elements [50, 51]. The atomic scale composition distribution in Fig. 10c-c₆ shows a random pattern in the atomic occupancy of each element in the outer layer of the nano rare earth inclusion.

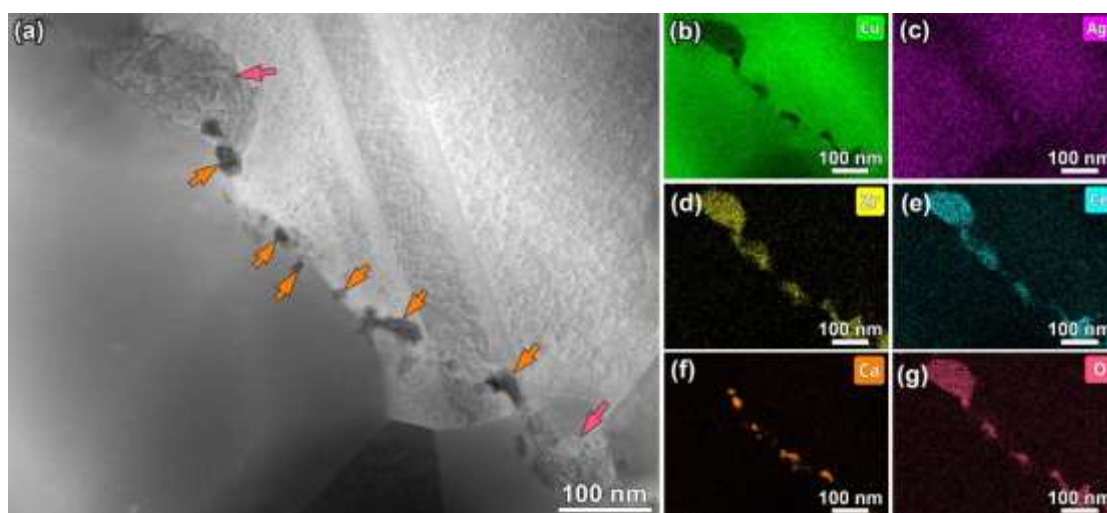


Fig. 9 (a) HAADF-STEM images and (b-g) corresponding EDS maps of the nano-sized rare earth oxides in aged Ce-containing alloy.

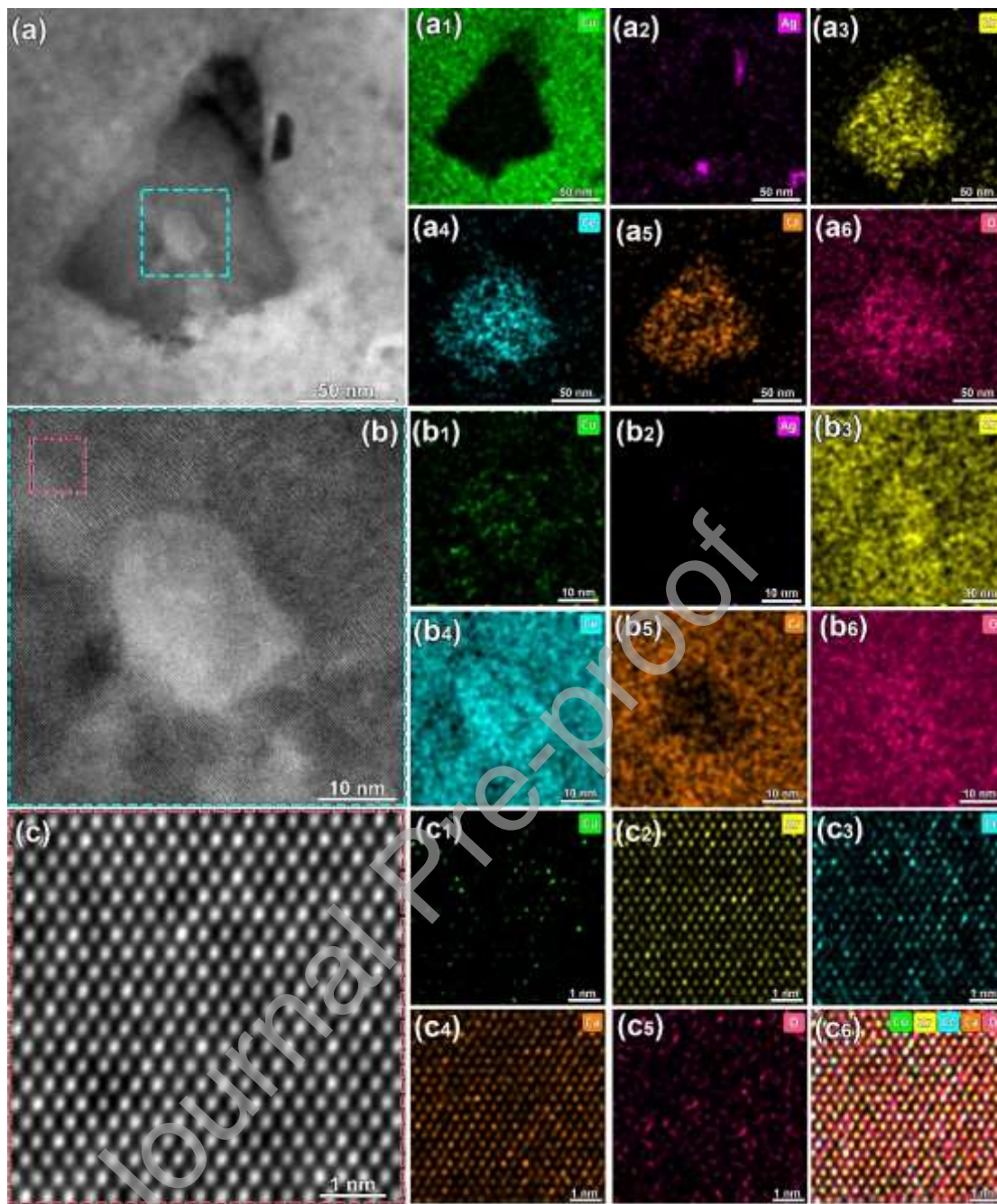


Fig. 10 (a, b) HAADF-STEM images and (a₁-a₆, b₁-b₆) corresponding EDS maps of the rare earth compounds with nano core-shell structure in aged Ce-containing alloy. (c-c₆) The atomic scale composition distribution in the outer layer of the nano rare earth compounds from the dashed box in figure 10b.

3.4 *In-situ* heating induced Ag precipitation

The tendency for Ag precipitates in Cu–Ag–Zr alloys to coarsen at service temperatures (500-

600 °C) is well-known, leading to a decrease in the alloy's high-temperature mechanical performance [12]. To investigate the impact of a trace amount of Ce on the precipitation and coarsening of the Ag phase, a method involving in-situ heating at 600 °C is employed to observe the evolution of Ag precipitates in two alloys continuously under a scanning electron microscope, as shown in Fig. 11. After a 30 min exposure to 600 °C, a significant quantity of precipitates emerges in the Ce-free alloy, as illustrated in Fig. 11a-d. With an increase in holding time, the number of precipitated phases increases and their size also grows larger. A magnified image of the precipitates in the Ce-free alloy after being held at 600 °C for 120 min is depicted in Fig. 11e. These precipitates display an irregular shape, with the largest size reaching about 500 nm. The EDS energy spectrum distribution map shows that these particles are Ag precipitates. In Cu-Ag alloy systems, this specific precipitation behavior is often observed, exhibiting discontinuous precipitation, significant size, and an incoherent relationship with the matrix [52-54]. The discontinuous Ag precipitates have a weaker strengthening effect on the matrix compared to the continuous Ag precipitates, leading to limited improvement in thermal and electrical conductivity of the material [26]. It is generally important to prevent the discontinuous precipitation of Ag during the service period. As shown in Fig. 11 f-j, only a small amount of discontinuous Ag precipitates in the Ce-containing alloy after heating at 600 °C for 30 min, with sizes not exceeding 100 nm. Extending the insulation time further, there is no significant increase in the quantity of discontinuous Ag precipitates, and the sizes do not significantly increase either. It can be seen from this that the addition of a small amount of Ce can effectively suppress the discontinuous precipitation and coarsening of Ag.

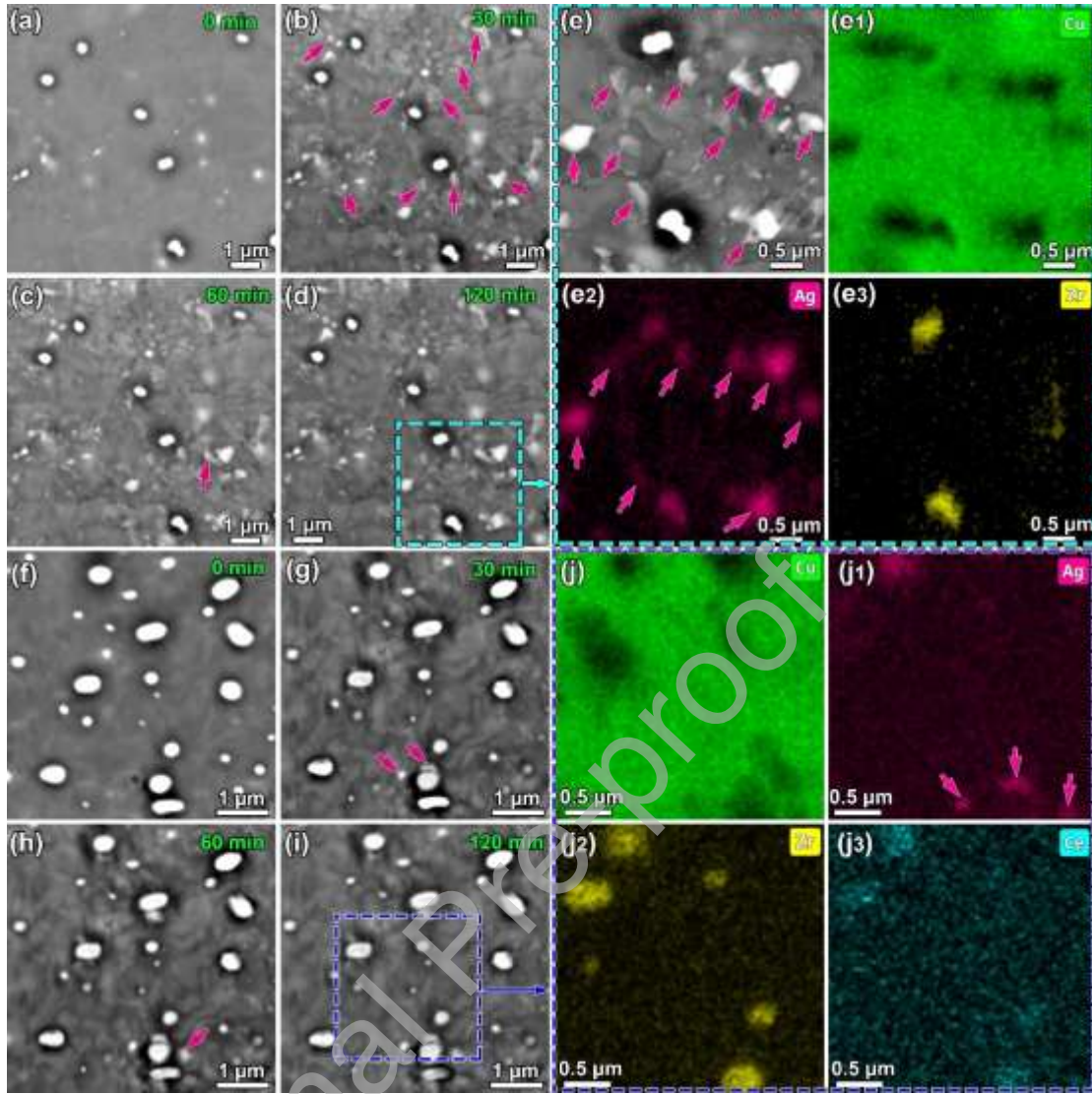


Fig. 11 In-situ SEM maps of rolled (a-d) Ce-free alloy and (f-i) Ce-containing alloy exposed to 600 °C at different time. (e) detail view from the dotted box in Fig. 11d, and (e₁-e₃) corresponding EDS mapping. (j-j₃) EDS mapping from the dotted box in Fig. 11i.

3.5 *In-situ* heating induced grain coarsening

The discontinuous precipitation and coarsening of the Ag phase at elevated temperatures are key factors responsible for the deterioration of the alloy's thermal stability. Additionally, grain coarsening under high-temperature conditions significantly contributes to the reduction in mechanical strength. Figure 12 a-g shows the continuous evolution process of the grains in the Ce-

free alloy at different heating times at 500 °C and 600 °C. Before in-situ heating, the aged Ce-free alloy retains a considerable amount of deformation structure, as shown in Fig. 12a. This deformed microstructural area persists throughout the entire in-situ heating process. In the blue deformed grain region (near (111) orientation), our focus is on the larger-sized A grain and the B grain clusters composed of numerous small fragmented grains. Both A grain and B grain clusters are oriented near (001) orientation. In addition, we also analyzed the small C grains (near (113) orientation) and D grains (near (314) orientation) outside the blue deformed grain region in the aged Ce-free alloy. The A grain exhibits a continuous slow change during the heating process. The top right corner of the grain undergoes an orientation change first, with the grain boundary gradually shrinking inward, being engulfed by the blue deformed grain region. The B grain clusters exhibit extremely high stability throughout the entire heating process. After holding at 500 °C for 15 min, some grains within the B grain clusters showed a trend towards transitioning from a near (001) orientation to the (101) direction (Fig. 12b). This trend became more pronounced with increasing holding time. Even after being kept at 500 °C for 240 min, and even continuing to be kept at 600 °C for 60 min, the grain size of the B grain clusters did not show significant changes (Fig. 12g). The C grain outside the blue deformation zone showed high stability during the 500 °C holding process, with no significant changes in grain orientation and size (Fig. 12a-e). However, after holding at 600 °C for 60 min, the grains began to significantly grow towards the blue deformation zone (Fig. 12g). The D grain exhibits instability in the early stages of insulation at 500 °C, and it rapidly grows after 30 min of insulation at this temperature (Fig. 12a-c). The severe migration of grain boundaries of the D grain leads to the formation of annealing twins in the interior. Finally, the D grain undergoes abnormal growth with prolonged insulation time and increased temperature (Fig. 12f, g). The Ce-

free alloy shows discontinuous recrystallization at the beginning of heating, then experiences localized grain coarsening after 30 min of heating.

As shown in Fig. 12h, the deformed grain region (indicated by a dashed box) is surrounded by the fine grain region in the aged Ce-containing alloy. With increasing heating time at 500 °C, the large area of the deformed grain region is continuously engulfed by the fine grain region, but the grains in the fine grain region do not show significant coarsening behavior (Fig. 12i-l). After heating at 600 °C for 30 min, the coarse deformed grain region has disappeared, leaving only fine equiaxed grains in the alloy (Fig. 12m). Furthermore, there is no significant coarsening of the equiaxed grains, nor any occurrence of abnormally large grains after heating at 600 °C for 60 min (Fig. 12n). It is evident that the Ce-containing alloy exhibits a tendency toward continuous recrystallization at 500 °C, leading to a uniform and fine recrystallized microstructure. Even when the temperature is increased to 600 °C during continuous annealing, no grain coarsening or abnormal grain growth is observed.

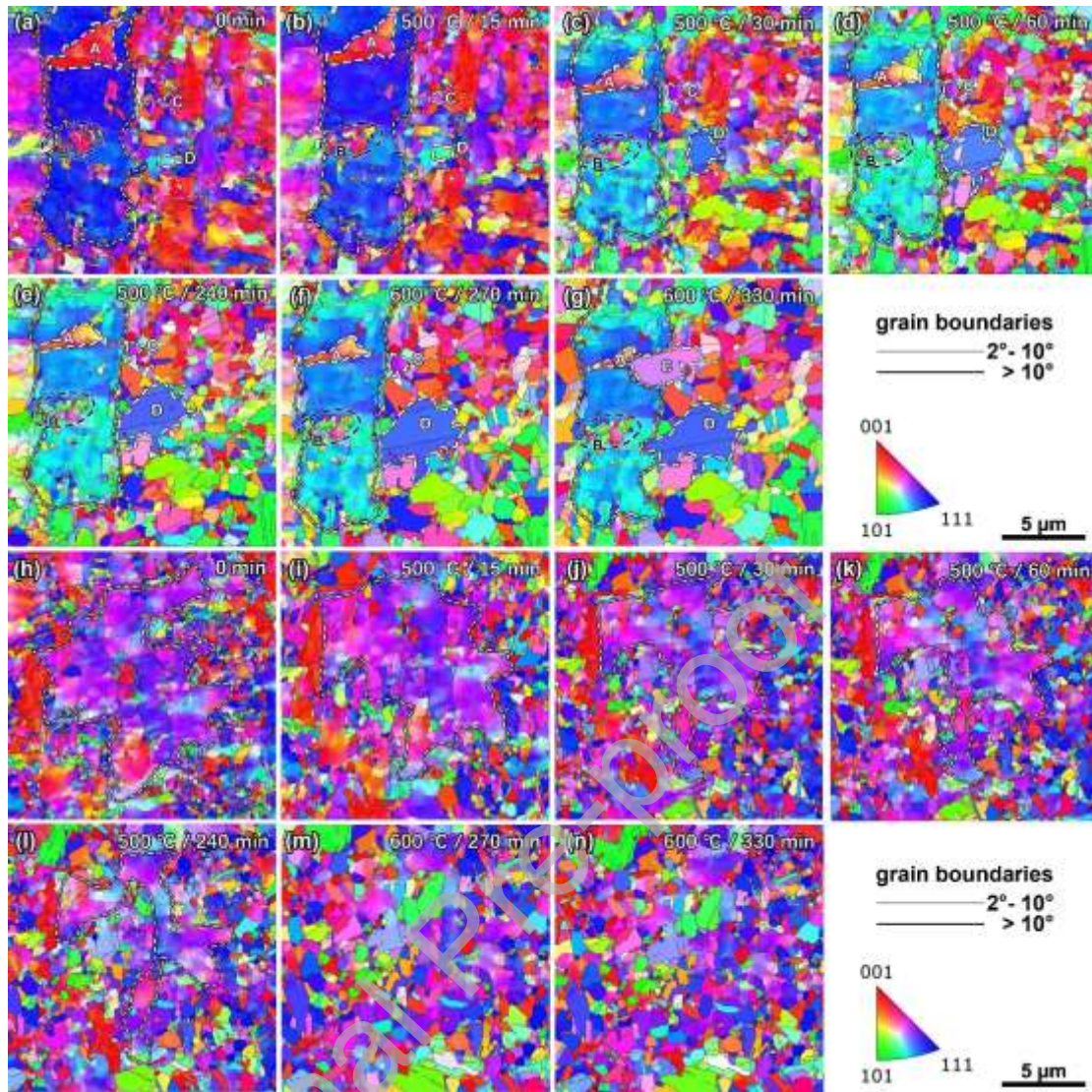


Fig. 12 In-situ EBSD-IPF maps of aged (a-g) Ce-free alloy and (h-n) Ce-containing alloy at different time and temperature.

Based on the recrystallization maps in Fig. 13, it is evident that the Ce-free alloy exhibited inhomogeneous recrystallization during heating, while the Ce-containing alloy underwent uniform recrystallization. Figure 14 illustrates the proportion of recrystallized grains and the average grain size changes in Ce-free and Ce-containing alloys at different heating stages. Before heating, the recrystallization degrees in both aged alloys are essentially the same. This is because the aging temperature (400 °C) is relatively low, causing only slow partial recrystallization in the two alloys.

However, the setting temperature of in-situ heating is higher than the aging temperature of most copper alloys, leading to a significant difference in the recrystallization and grain growth behavior of the two alloys. In the early stages of in-situ heating experiments, both alloys began to undergo recrystallization, with a sharp decrease in average grain size. However, the Ce-containing alloy exhibited a significantly faster recrystallization nucleation rate. As the holding time extends to 60 min, the proportion of recrystallized grains in the two alloys reaches around 40%. At this point, the recrystallization rate slows down, and the grain refinement process almost stagnates. When the temperature reaches 600 °C, the residual deformed grains in both alloys undergo recrystallization. The average grain size continues to decrease in the alloy with added Ce. However, the grains that have already undergone recrystallization continue to grow in the Ce-free alloy, causing abnormal grain growth and resulting in an increase in the average grain size of this alloy. It is not difficult to see that adding Ce makes recrystallization nucleation easier, but effectively inhibits the grain growth process.

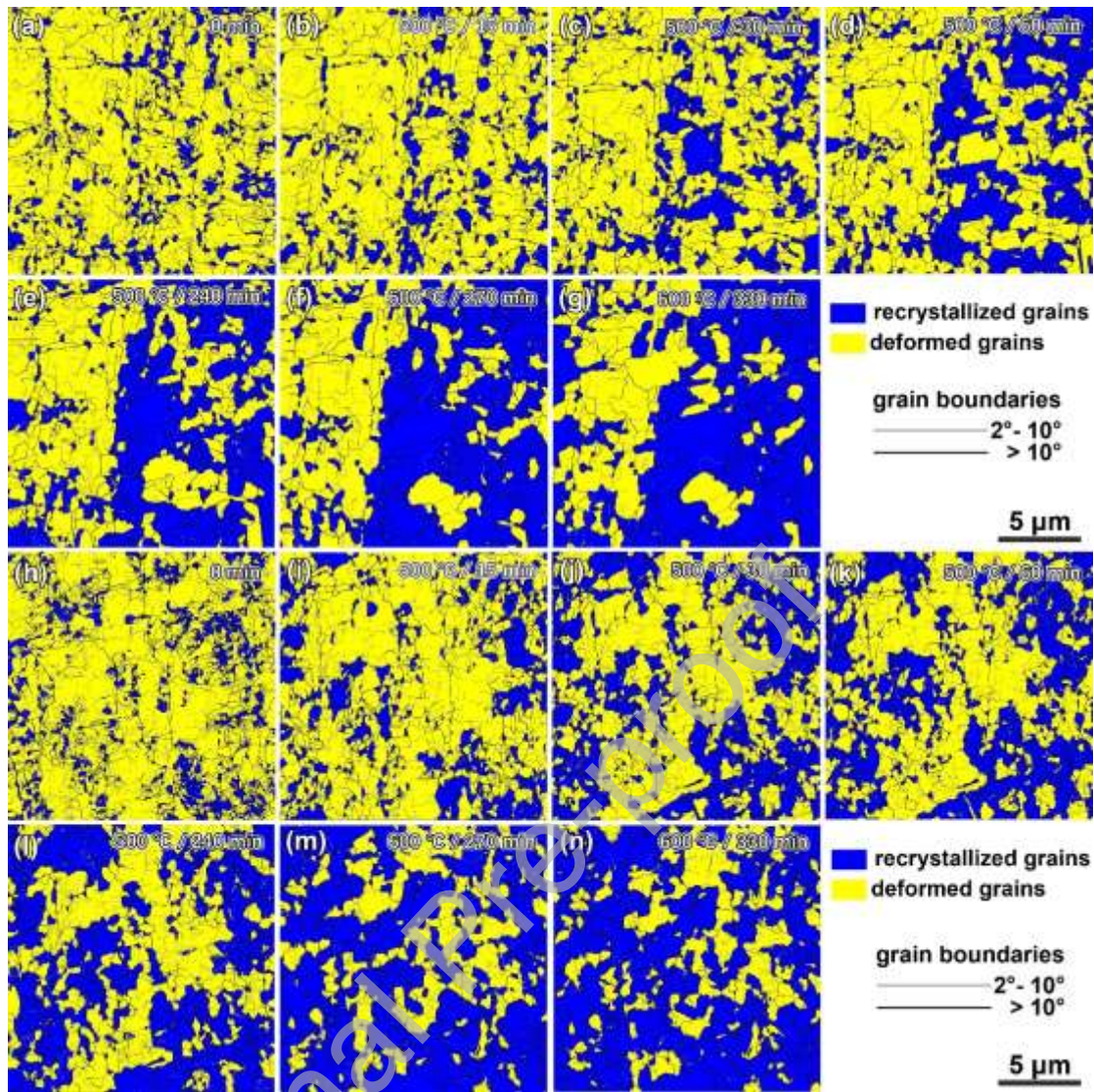


Fig. 13 In-situ EBSD-recrystallization maps of aged (a-g) Ce-free alloy and (h-n) Ce-containing alloy at different time and temperature.

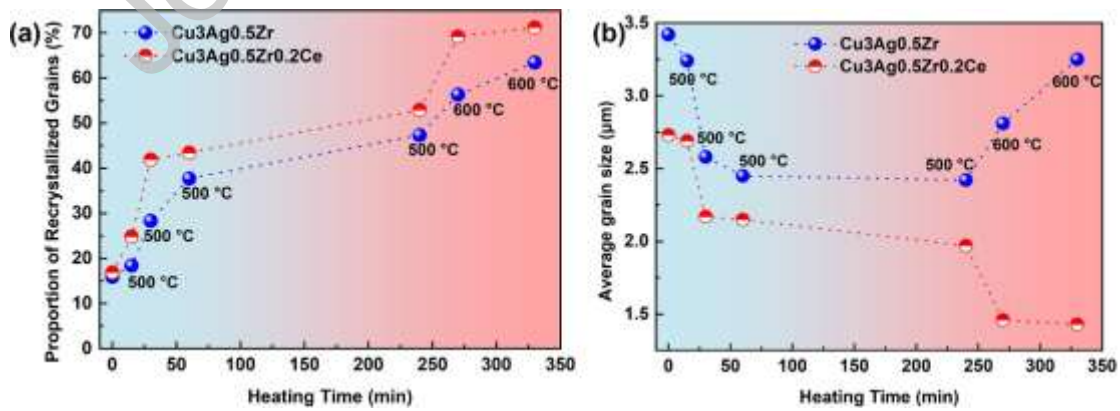


Fig. 14 The evolution of (a) proportion of recrystallized grains (b) and the average grain size in the

Ce-free and Ce-containing alloys during the in-situ heating process.

4 Discussion

4.1 Role of Ce microalloying in purifying the Cu matrix

The rare earth element Ce serves as an effective modifier for Cu–Ag–Zr alloy, achieving dual purification of both the matrix and grain boundaries through selective adsorption and immobilization of harmful impurities. The XRD analysis in Fig. S3 indicates that the addition of Ce induces a slight rightward shift in the Cu diffraction peak. In the Ce-free alloy, impurity elements such as Ca within the Cu face-centered cubic lattice induce lattice distortion, resulting in an increased interplanar spacing (d) and a decreased diffraction angle (θ), which causes the Cu diffraction peak to shift leftward relative to that of pure copper. Upon the introduction of trace amounts of Ce, impurity elements are effectively scavenged from the Cu lattice, thereby alleviating lattice distortion. Consequently, the interplanar spacing (d) decreases and the diffraction angle (θ) increases, leading to a slight rightward shift in the Cu diffraction peak compared to the Ce-free alloy.

The purification mechanism of Ce in copper alloys is primarily attributed to differences in electronegativity and atomic-scale interfacial behavior. Ce exhibits strong chemical selectivity due to its electronegativity of 1.12, which is notably lower than that of Zr (1.33), Cu (1.90) and O (3.44) but marginally higher than Ca (1.00) [55]. With its higher affinity for oxygen compared to copper, cerium can selectively react with non-metallic impurities. As depicted in Fig 9, this reaction leads to the production of thermally stable rare earth oxides, effectively preventing the formation of detrimental inclusions like Cu_2O . Moreover, the atomic radius of Ce (0.182 nm) and Cu (0.128 nm) exhibit a significant difference, limiting its solubility in the copper matrix. As a result, Ce tends to segregate at high energy regions of grain boundaries, where it adsorbs O and Ca impurities, reduces

electron scattering centers, and forms fine spherical oxides below 100 nm in size. This process eliminates lattice distortions induced by impurities and effectively removes harmful elements from grain boundaries, enabling comprehensive purification. The uniform dispersion of rare earth oxides introduces minimal obstruction to electron transport, thereby preserving high thermal conductivity at elevated temperatures and avoiding the degradation typically caused by conventional impurities.

On the other hand, the addition of trace amounts of rare earth Ce to the alloy enhances high-temperature plasticity and strength through grain boundary purification and dispersion strengthening mediated by nano-sized rare earth oxide particles. Cerium reacts with impurities such as oxygen and calcium in the alloy, forming thermally stable rare earth oxides that immobilize these impurities and prevent embrittlement caused by their segregation at grain boundaries. The dispersed nano-sized rare earth oxide particles effectively hinder dislocation motion, serve as pinning sites to suppress grain growth, and retard dynamic recovery and recrystallization processes, thus contributing to enhanced high-temperature strength and ductility[22, 56]. This synergistic combination of grain boundary purification and dispersion strengthening results in significantly improved high-temperature mechanical performance in copper alloys.

4.2 Atomic Ce substitution induces precipitate refinement

The size, quantity, uniform distribution, and thermal stability of second phase particles play crucial roles in determining the high-temperature microstructural stability of alloys, which subsequently impacts their mechanical performance retention [57, 58]. Ideally, when the alloy exhibits a finer second phase particle size, increased quantity, well-dispersed distribution, and enhanced thermal stability, it tends to demonstrate superior high-temperature mechanical properties[14, 16, 17, 42, 43]. The high-temperature mechanical properties of the alloy are greatly

improved by the addition of Ce, primarily through its influence on the formation of the Cu_4AgZr phase and the precipitation of Ag.

As depicted in Fig. 11, the Ce-free alloy shows minimal coarsening of the submicron Cu_4AgZr phases, which is vital for pinning grain boundaries and preventing grain growth, thereby improving the alloy's high-temperature strength [8]. This means that enhancing thermal stability in the Cu-Ag-Zr alloy could be achieved by creating smaller, more numerous, and better dispersed Cu_4AgZr particle phases. By adding minor concentrations of Ce, the volume fraction of the $\text{Cu}_4\text{Ag}(\text{ZrCe})$ phase in the alloy is enhanced without enlarging the phase, as evidenced by the statistical analysis depicted in Fig. 8 and Table 4. In the Ce-free alloy, only 24% of the Cu_4AgZr phase sizes are smaller than 200 nm. Conversely, the Ce-containing alloy exhibits a substantial 68% of the $\text{Cu}_4\text{Ag}(\text{ZrCe})$ phase size below 200 nm with identical plastic deformation and aging processes. During the aging process, a considerable number of $\text{Cu}_4\text{Ag}(\text{ZrCe})$ phases smaller than 200 nm are thought to form in Ce-containing alloys, as shown in Fig. 5. While the precipitation of Cu_4AgZr phases of this size is rarely observed during aging in the Ce-free alloy. Moreover, the Ce-free alloy contains a volume fraction of 5.8% of the Cu_4AgZr phase. Upon the addition of rare earth element Ce, the volume fraction of this phase in the alloy increases to 7.1%. Adding Ce is advantageous for promoting the nucleation of the $\text{Cu}_4\text{Ag}(\text{ZrCe})$ phase, leading to an increase in the volume fraction of this stable phase in the alloy. Based on the atomic-scale compositional analysis in Fig. 4, the occupancies of Cu and Ag atoms remain unchanged in the $\text{Cu}_4\text{Ag}(\text{ZrCe})$ phase compared to the Cu_4AgZr phase. However, Ce atoms partially substitute for Zr at the Zr lattice sites. This indicates that Ce and Zr compete for occupancy in the same crystallographic positions during the formation of this type of secondary phase. Introducing a more reactive rare earth element can significantly enhance the

nucleation rate of the second phase while hindering its coarsening [59, 60]. A similar competitive mechanism has been identified in Cu-Ni-Co-Si alloys [61, 62].

Our team initially explored increasing the Zr content in the alloy to enhance thermal stability before considering adding trace rare earth elements. The aim is to increase the volume fraction of the submicron Cu_4AgZr phase in order to enhance the high-temperature strength and plasticity of the Cu-3Ag-0.5Zr alloy. Increasing the Zr content to 0.8% results in a significant rise in the volume fraction of the submicron Cu_4AgZr phase, as depicted in Fig. S4 and Table S1. However, this causes coarsening and uneven distribution of the submicron phase particles, leading to a slight decrease in strength and plasticity of the Cu-3Ag-0.8Zr alloy at high temperatures. The incorporation of the rare earth element Ce not only improves the volume fraction of submicron thermal stable phase particles in the alloy but also ensures their uniform distribution and prevents their coarsening. This synergistically enhances the alloy's high-temperature strength and plasticity. Furthermore, comparing Fig. 5 and Fig. S4 shows that incorporating trace Ce is more beneficial in spheroidizing the submicron second phase particles. This spheroidization aids in minimizing stress concentration during alloy loading, leading to increased strength and plasticity at elevated temperatures.

Besides, enhancing the high-temperature mechanical performance of the alloy can be achieved by optimizing the size and volume fraction of Ag precipitates in the alloy, as well as ensuring its stability during the heating process. The addition of Ce can effectively inhibit the discontinuous precipitation and coarsening behavior of Ag. In the Ce-free alloy, extensive precipitation of Ag-rich phases within the grains and at the grain boundaries with sizes ranging from tens to over one hundred nanometers is observed when aged at 400 °C (Fig. 5). Most of these larger precipitates show an incoherent interface with the Cu matrix. When heated to high temperatures (600 °C) in situ,

discontinuous Ag precipitates in Ce-free alloys can exceed 500 nm in size (Fig. 11). The presence of coarse and discontinuous Ag precipitates is a clear indication of reduced thermal stability in the alloy. As summarized in Table 4, the discontinuous Ag precipitates have a volume fraction of 4.6% with the average diameter 73 nm in the Ce-free alloy. When subjected to the same heat treatment aging process, the addition of rare earth Ce causes the size of the Ag precipitates to decrease to one twenty-sixth of their original size, while their volume fraction increases by more than one time, as displayed in Fig. 8 and Table 4. Furthermore, these finely dispersed Ag precipitates within the grains are coherent with the matrix, and no discontinuous Ag precipitates are observed (Fig. 7). The addition of Ce promotes the formation of a greater volume fraction of the $\text{Cu}_4\text{Ag}(\text{ZrCe})$ particles by sequestering more Ag atoms into this thermally stable intermetallic, thereby suppressing Ag coarsening at the source. Moreover, rare earth solutes retard Ag diffusion during thermal exposure, inhibiting both discontinuous precipitation and coarsening of Ag [26, 34, 39, 45]. This means that the addition of Ce can enhance the stability of the interface between Ag precipitates and the matrix. This, in turn, prevents the dissolution and coarsening of the fine Ag, strengthens the pinning effect on dislocations and grain boundaries, and ultimately suppresses rapid grain boundary migration. Zr is widely recognized for its ability to prevent the discontinuous precipitation of Ag [4, 6, 25]. By increasing the Zr content in the alloy to 0.8%, as shown in Fig. S4 and Table S1, the coarsening of Ag phases during the aging process is effectively suppressed. However, the refining effect of the rare earth Ce is more noticeable.

4.3 Role of Ce microalloying on the recrystallization behavior

The presence of Ce promotes recrystallization nucleation, refines grain size, inhibits grain coarsening, and enhances the thermal stability of grains according to the in-situ heating EBSD

experiment result, as shown in Fig. 12 and Fig. 13. The alloys retain the stored strain energy in their internal structures after experiencing cold deformation, and this stored strain energy, along with surface energy, serves as the driving force for recrystallization [56]. It is noteworthy that, for a given level of cold deformation applied to both alloys, the volume fraction of submicron/micron second phase particles exerts a pronounced influence on the stored strain energy[58]. The submicron/micron second phase particles collide and crush with the matrix during rolling, leading to the accumulation of significant localized strain energy[8, 58]. Adding a small amount of Ce to the alloy results in a significant increase in the volume fraction of submicron particles, which in turn improves the strain energy and enhances the driving force for subsequent recrystallization. On the other hand, the submicron $\text{Cu}_4\text{AgZr}/\text{Cu}_4\text{Ag}(\text{ZrCe})$ particles will initiate recrystallization nucleation through the particle stimulated nucleation (PSN) mechanism, which is well known and widely observed in Mg alloys [63, 64], zinc alloys [65], and aluminum alloys [66]. By incorporating rare earth elements as nanoscale precipitates in the copper alloy, such as nanoscale Cu_6La [56], and Cu_4Sc particles [22], it is possible to effectively prevent recrystallization nucleation and grain boundary migration. The retarding effect is attributed to the homogenization of dislocation distribution by the nanoscale particles, reducing the orientation gradient required for nucleation and consequently inhibiting nucleation [57, 58]. Conversely, incorporating rare earth elements as submicron/micron second phase particles tends to facilitate recrystallization through the PSN mechanism in the copper alloy [35, 67]. The submicron $\text{Cu}_4\text{AgZr}/\text{Cu}_4\text{Ag}(\text{ZrCe})$ phase particles typically act as preferential nucleation sites for recrystallization and play a crucial role in initiating recrystallization nucleation in this study. The number and placement of recrystallization nuclei are determined by the amount and distribution of these second phase particles. The uneven distribution

of Cu₄AgZr particles with a relatively smaller quantity in the Ce-free alloy results in non-uniform recrystallization PSN nuclei during heating. Adding a small amount of Ce leads to the formation of a greater quantity of submicron Cu₄Ag(ZrCe) particles in the alloy. The high density of Cu₄Ag(ZrCe) particles is uniformly dispersed throughout the matrix, facilitating rapid and uniform recrystallization PSN nuclei during heating.

The addition of Ce not only promotes recrystallization nucleation in the alloy but also hinders the growth of recrystallized grains. The second phase particles exert a strong pinning effect (Zener pinning) on grain boundaries, with the pinning pressure being determined primarily by the size, volume fraction, distribution and stability of the particles[58]. The smaller the size of the second phase particles, the higher their volume fraction, the more uniform their distribution, and the better their thermal stability, leading to a stronger pinning force at grain boundaries, increased resistance to grain boundary migration, a slower grain growth rate, and a more significant suppression of grain coarsening [57, 58]. Hence, concerning the size and volume fraction of the second phase, nanoscale Ag precipitates demonstrate a more pronounced inhibitory impact on grain boundary migration compared to the Cu₄AgZr/Cu₄Ag(ZrCe) phase. The stronger pinning ability on grain boundaries and more effective suppression of grain growth is indicated by the smaller dimensions and higher volume fractions of nano-sized Ag precipitates. The addition of Ce effectively suppresses discontinuous precipitation and coarsening of Ag precipitates. The Ag precipitates exhibit a uniform nanoscale morphology with an average diameter of 2.8 nm in the Ce-containing alloy, making up 10.2% of the volume. While the size of nano Ag particles grows to 72.9 nm, and the volume fraction drops to just 4.6% in the Ce-free alloy. Additionally, it has been observed that nano-sized Ag precipitates in Ce-free alloys tends to coarsen when heated at 600 °C for 0.5 h (Fig. 11). However,

there is no significant coarsening of Ag observed in the alloy containing Ce even after being held at 600 °C for 2 h. The addition of Ce significantly enhances the thermal stability of nano-sized Ag precipitates, consequently improving the high-temperature mechanical properties of the alloy. Furthermore, in-situ heating experiments have demonstrated the exceptional stability of the submicron $\text{Cu}_4\text{AgZr}/\text{Cu}_4\text{Ag}(\text{ZrCe})$ particles, as they do not exhibit any coarsening when exposed to 600 °C for 2 h. Consequently, the effectiveness of these precipitates in impeding grain boundaries during grain growth mainly depends on their original size, amount, and dispersion. As displayed in Fig. 5 and Table 4, adding Ce not only increases the volume fraction of the submicron particles but also effectively refines them, which is beneficial for enhancing their inhibitory effect on grain growth. This will improve the alloy's thermal stability at high temperatures.

5 Conclusions

A novel copper alloy, reinforced by minor Ce, is designed to exhibit good thermal stability. The addition of trace amounts of Ce effectively enhances the high-temperature strength and plasticity, without significantly compromising the high-temperature thermal conductivity of the alloy. The influence of the rare earth Ce on the stability of the second phase and recrystallization behavior during heating is analyzed by in-situ high temperature SEM/EBSD. Conclusions drawn are as follows:

- (1) The addition of Ce helps to solidify free oxygen and impurity elements within the copper matrix, thus enhancing the high-temperature strength and plasticity of the alloy
- (2) The addition of Ce increases the population of submicron secondary phase particles and refines them at the same time. These particles act as efficient nucleation sites for recrystallization, leading to a considerable reduction in the final grain size.

(3) Ce suppresses both discontinuous precipitation and coarsening of nanoscale Ag precipitates, strengthening the Zener pinning effect of the secondary phase on grain boundaries. By inhibiting grain coarsening during recrystallization, Ce markedly enhances the alloy's high-temperature mechanical properties.

(4) Adding trace rare earth Ce can enhance the strength and ductility of the Cu-Ag-Zr alloy at high temperatures, with minimal impact on its thermal conductivity at 500 °C (only a slight decrease of approximately 1.5%).

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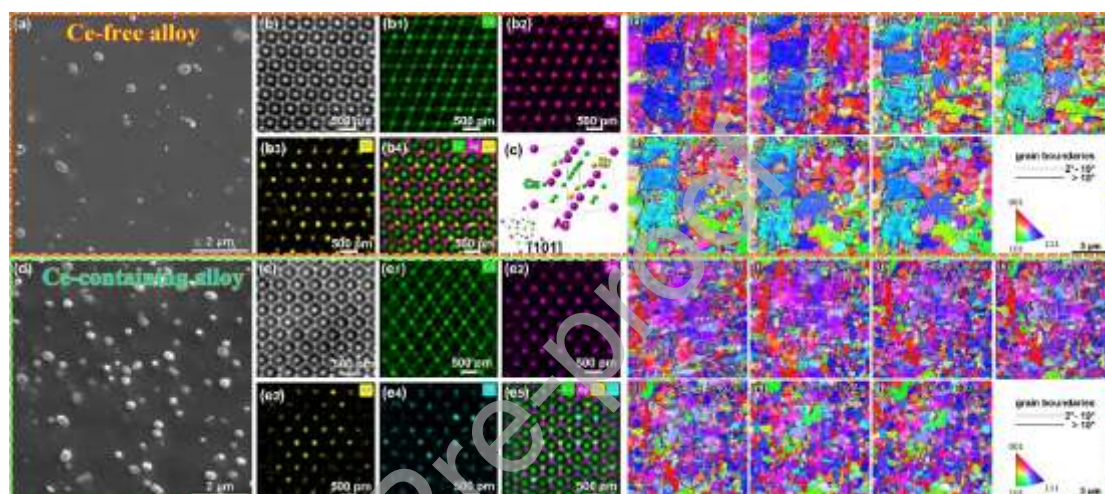
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Graphical Abstract



Declaration of Interest Statement

No conflict of interest exists in this paper and the paper is approved by all authors for publication. I would like to declare that the work described in this paper is our original research that has not been published in any form previously, and not under consideration for publication elsewhere, in whole or in part. All the authors listed in this paper have directly participated in planning, execution, and analysis of this study.